Wireless Sensor Networks: P2 Exercises **13/10/2026**

For the following exercises create a new C source file per exercise (for example: p2_ex1.c, p2_ex2.c), based on the procedure followed in the lab examples (remember the modifications to be performed in the *makefile* and the properties of the project to successfully compile and upload the code into the Cookie node).

Based on that, proceed with the following exercises:

1. Use the serial terminal as a command line for the Cookie to execute actions, such as toggle a LED and request a measurement from a specific sensor. For that purpose, a frame format must be defined to specify the type of action and the sensor/actuator to be triggered. For example, consider the following examples:
    a. "LR" or "LG" means action on the LEDs (L) and toggle the LED Red (R) or Green (G), which corresponds to one of the LEDs in the Cookie.
    b. "ST" or "SH" means action on the sensors (S) and request a measurement from the Temperature (T) or Humidity (H) sensors.

    Create a process to manage the incoming messages from the command line, and other processes to manage the sensors and actuators (which will be activated based on decoding the incoming messages in the first process).
    The actions will be performed under request, which means that the sensing or actuation is performed just when it is required.

2. As a result of exercise 1, use the command line to also configure additional parameters of the node. Another field in the frame format can be included to configure, for instance, the time interval of the actions to be performed. Consider as an example the case of the temperature sensor ("ST"). Including an additional character for the periodicity of the data (e.g. "ST2") means that the measurement is to be performed every 2 seconds. If no value is specified or the value is 0, it means only one measurement. Proceed similarly with the rest of the cases.
    (**NOTE**: Check in the timer library how to reset an *etimer* with a new time interval).

Evaluation

The exercises will be evaluated through a document that should include the following items:

- Code of each exercise (inside the document).
- Brief explanation with screen captures showing the results of the exercise.
- Upload this document to **Moodle** by Friday the **23th of October** (a specific homework task will be created in Moodle for it).
- Remember this is part of the evaluation of the subject.